# FPGA-Assisted Portable Offline Speech Translation Device using Raspberry Pi and PYNQ Board


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| **Group No: 54** | |

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**Preliminary Study:**

Communication plays an important role in connecting people and sharing information. However, language differences often create difficulties when people from different linguistic backgrounds interact with each other. This issue is commonly faced in situations such as travelling, healthcare, education, business communication, and emergency scenarios, where understanding each other quickly is important.

In recent years, several language translation applications have been developed to reduce language barriers. Mobile applications provide features such as text translation, speech recognition, and voice-based translation. However, most of these applications depend on internet connectivity and cloud-based processing. This makes them less reliable in remote locations where internet access is limited. Additionally, using a smartphone-based solution may not always be convenient in situations where a dedicated and portable communication device is required.

With the development of embedded systems, artificial intelligence, and hardware acceleration technologies, it has become possible to create compact devices capable of performing intelligent tasks efficiently. This creates an opportunity to develop a portable speech translation system that can provide quick and reliable communication assistance without complete dependence on external services.

The proposed project aims to develop an AI-based portable speech translation device using Raspberry Pi and PYNQ FPGA board. The system is designed to capture speech from a user through a microphone, process the input, translate it into a selected target language, and provide the translated output through a speaker. The device will focus on supporting multiple languages, enabling easier communication between individuals who do not share a common language.

The Raspberry Pi will act as the main processing and control unit of the system, handling user interaction, speech processing workflow, and communication between different components. The PYNQ FPGA board will be integrated to explore hardware acceleration and improve the efficiency of computationally intensive operations involved in speech processing. This combination of software-based intelligence and hardware acceleration will allow the project to demonstrate the application of AI on embedded platforms.

The system will incorporate a push-to-talk mechanism, allowing users to activate speech recording only when required. This approach helps in reducing unnecessary processing, improving power efficiency, and

making the device more practical for real-world usage. The project aims to create a low-cost, portable, and user-friendly translation device that can assist people in overcoming language barriers.

This project combines concepts from artificial intelligence, embedded systems, digital signal processing, and FPGA-based computing. The final objective is to develop a working prototype that demonstrates the possibility of implementing multilingual speech translation on a compact hardware platform.

**Objectives of the Project work:**

- Develop a portable offline speech-to-speech language translation system using the PYNQ FPGA board.
- Capture speech input through a USB microphone connected to the PYNQ board.
- Perform FPGA-accelerated audio preprocessing and noise reduction.
- Convert speech to text using an offline speech recognition engine running on the PYNQ board.
- Translate the recognized text between selected languages without requiring an internet connection.
- Convert the translated text into natural speech using an offline text-to-speech engine.
- Minimize translation latency through hardware/software co-design and FPGA acceleration.
- Design an efficient embedded architecture integrating FPGA logic and ARM processing on the PYNQ platform.
- Demonstrate real-time communication from the selected languages in offline and low-connectivity environments.

**Component and Software required to carry out the project:**

| **Component** | **Quantity** |
|---|---|
| Raspberry Pi 3B+ | 1 |
| PYNQ-Z2 FPGA Board | 1 |
| USB Microphone | 1 |
| USB Speaker | 1 |
| MicroSD Card (32 GB) | 2 |
| Ethernet Cable | 1 |
| Power Adapter (5V) | 2 |
| Jumper Wires | As Required |
| Breadboard (Optional) | 1 |
| OLED Display (Optional) | 1 |

**Cost estimation of the project:**

| Component | Estimated Cost (₹) |
|---|---|
| Raspberry Pi 3B+ | 5,000 |
| PYNQ-Z2 FPGA Board | 18,000 |
| USB Microphone | 900 |
| USB Speaker | 700 |
| MicroSD Cards | 1,200 |
| Ethernet Cable | 300 |
| Power Adapters | 1,000 |
| Miscellaneous | 1,400 |
| Total Estimated Cost | 29,000 (round off) |

**Work distribution among the project group members:**

## Ananya Singh

- Literature survey and analysis of existing speech translation systems.
- Research paper review related to AI translation, embedded systems, and FPGA applications.
- Documentation management throughout the project development.
- Assistance in testing, result analysis, and final report preparation.

## Pranjali Sinha

- Raspberry Pi environment setup and system configuration.
- Implementation and integration of Speech-to-Text (STT) module.
- Development and integration of the language translation pipeline.
- Software testing and performance evaluation.

## Utkarsh Raj

- FPGA development using Vivado and PYNQ framework.
- Design and implementation of PYNQ overlays.
- FPGA-based DSP/audio preprocessing module development.
- Raspberry Pi–PYNQ communication and hardware-software integration.
- System optimization and hardware acceleration analysis.

## Yash Panhale

- Text-to-Speech (TTS) module development and integration.
- User interface/GUI development for system interaction.
- Hardware assembly, peripheral interfacing, and prototype setup.
- System demonstration, validation, and performance testing.

| S.No. | Name of Supervisor | Signature of Supervisor |
|---|---|---|
| 1. | Prof. Tejaswini Kar | |

| S.No. | Name of Students | Signature of Students |
|---|---|---|
| 1. | Ananya Singh | |
| 2. | Pranjali Sinha | |
| 3. | Utkarsh Raj | |
| 4. | Yash Panhale | |